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数学建模国际赛

承 诺 书

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参赛队教练员（签名）： 周长银

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Terrorism Evaluation and Classification based on Big data

Abstract:

Based on the conditions and requirements of this problem, under reasonable assumptions, we established two models. One is the TREI (S) model, and another is SKNN outlier classification model which is based on the K-nearest neighbor algorithm, using big data processing method. Through these two models, you can get risk index, evaluating monitored person's situation of using network, and can classify large amounts of data effectively, quickly and automatically.

For task 1, we establish the TREI (S) model consists of the social attributes and network attributes. The individual risk index can be evaluated according to the weights given to different attributes. Then we can score their social attributes by the real information of the monitored people. After classifying the network behavior of the monitored people, we will obtain the relation graph theory. Then we need to calculate the center of his degree and its centrality, and we will get its network centrality attribute score. The weighted sum of the two scores is the index of the individual risk. Compared with the basic model TREI, our model considers the social attributes, and can evaluate the individual terrorist threat more comprehensively. The model is verified by the data of the Paris case, and the results (*Tab. 19*) are in line with the actual situation of the criminals involved in the terrorist attacks in Paris.

For task 2, we establish an improved SKNN (Swift KNN) model. KNN algorithm has two disadvantages, one is the high computational complexity, the other is that, in the case of unbalanced sample size, it is easy to lead to that the majority of the neighbors belonging to the new sample are in the large capacity class. In this algorithm, we appropriately reduce the dimension of feature space, use a small sample library, and find the nearest neighbor in the shortest possible time. Data validation, as for the 800 samples' data, the calculation efficiency of the SKNN algorithm is about 23 times higher than that of KNN(*Tab. 23*), and the classification accuracy is very high. It is obvious to see that the classification SKNN model based on the K- nearest neighbor algorithm can classify the big data of Internet effectively, quickly and automatically.

Finally, we analyze and evaluate the model based on the analysis results. According to the results of the model, we put forward some reasonable opinions and suggestions concerning serious anti-terrorism problems to President Obama in the form of a letter.

Key words: big data processing, individual risk index, categorization

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I. Introduction

In order to indicate the origin of terrorism problems, the following background is worth mentioning.

1.1 Problem Background

Terrorism is the implementer of the non-armed organized use of violence or threats of violence, certain objects by placing terror to achieve certain political objectives behavior. The international community in some organizations or individuals to take kidnappings, assassinations, explosions, air hijacking, hostage-taking and other terrorist means, seek ideas and actions to achieve their political objectives or a specific requirement.^[1]

Contemporary terrorism began to take shape in the late 1960s, usually in a series of explosions academia, assassinations, kidnappings, hijackings and other terrorist incidents occurred in Europe in 1968 as the starting point of contemporary terrorism, and in the decades that followed the years, terrorist activities intensified, until the '9.11' incident reached a peak. Currently, the definition of terrorism, the more common definition is: Terrorism is the use of violence or threat of use of violence; terrorism has a clear political purpose, in order to produce an atmosphere of terror, usually harm innocent civilians objects: Terrorism is an organized activity systems typically have extraordinary resistance, anti-moral, arbitrary, concealment, uncertainty and unexpected features.^[2]

With the development of the network, terrorism has gradually developed a complex, subtle dynamic network model, and derive a series of new features, which makes for a more difficult fight against the terrorist organization. Unlike traditional networks, terrorist networks do not have a strict hierarchical structure, contact between members having arbitrary and hidden, so that terrorist networks showing loose, casual and covert organizational structure. Redundancy and flexibility of the network has become the main features of terrorism. In addition, even if the central node in the network of terror is removed, usually the network will not be destroyed. Because the redundancy and flexibility of terrorist networks so that it can quickly rebuild without losing its functionality. Moreover, concealment and lack of data also allows for information to combat terrorist networks become a problem.^[2]

The fight against terrorism is imminent. We need to use scientific methods, the use of scientific weapons to fight the inhuman terrorist jihad. According to the terrorists we monitor e-mail exchanges between the peer network behavior, we use data mining technology network communication diagram between them build out and take advantage of big data processing techniques to analyze these data, to grasp their movements, help us prevent and combat terrorist attacks. Moreover, we can also use our data in the hands of the collapse of their correspondence, provide support and help for the complete removal of a terrorist organization.

1.2 Previous Research

The analysis of the terrorism has been a universal concern. Moreover, there have been a lot of research into the studies of risk index. For the risk index and the data mining of suspicious person, which are considered as important factors in solving terrorism, there are a lot of previous research that can facilitate our understanding of the model.

For example, In the winter of 2001, "the international social network analysis network" (INSNA) specialized online journal "Connection" for the problem would benefit several network analysis and terrorism has been specifically discussed.^[3] Valdis E.Kreb is the first one to build the event of the 9/11 terrorist network model, his use of publicly available information to build the 19 hijackers were of personal relationships, and found a relationship between them.^[4] Nasrullah Memon et al., rebuilt this model in 2007, were factional, sectarian and K- cluster computing and characters between events analyzed by the model, found that 911 key people and key events in the subgroups.^[5] In 2007, D.M.Akbar Hussain proposes a new social network analysis methods to analyze the characters through a variety of attributes, building pluralistic property values for tasks, enabling a more comprehensive evaluation of the characteristics of a terrorist, and found a key figure. In the same year December, D.M.Akbar Hussain applied this model to 911 events of a terrorist organization, which found that a key figure in the terrorist network, and analyzed easily destroy terrorist networks and vulnerability.^[6]

1.3 Our Works

We are asked to build a realistic, effective, and useful model to solve terrorism. Our model should not only consider the monitored people's internet use and categorize the big data we get from the monitored people, but also other inessential factors that we need to consider, such as their social activities.

At first, we searched a big number of paper that studied the solution of terrorism and outliers to help us deepen the understanding of the problem. Valdis E.Kreb is the first one to build network model, and then Nasrullah Memon improved that model in 2007. Later, we found out that the traditional network model cannot evaluate the terrorists' risk index completely. A person educated, nurturing environment and religious factors will affect his future choice of values. So consider terrorists community property is a very necessary content. For example, many terrorist attacks are carried out in the name of religion, therefore investigate whether the monitored person's religious beliefs and terrorist organizations have countless ties to evaluate the person's risk index has a very important significance. Again, an unmarried staff required social responsibility and family responsibilities is less than a married person, the consequences of carrying out terrorist activities when he needs to be considered is small, and therefore more dangerous.

Based on the idea of big data processing have been obtained for data classification analysis and exception data processing, we selected KNN algorithms. KNN algorithm in 1968 proposed by Cover and Hart, is a relatively mature theoretical algorithms, and its working principle is: the existence of a set of sample data, and the data are present in each sample set label that we know the relationship between each data sample belongs classification. Enter the new data without a label, will each feature a new centralized data and sample data corresponding characteristics were compared, and algorithms to extract classification label features the most similar sample set of data. In general, we select only the sample data set before the k most similar data, which is the k - nearest neighbor algorithm source k , k is an integer of usually greater than 20. Finally, select the maximum number of classification k most similar data appear as a classification of new data. Those data, we call outliers, is to be classified out of the data. These anomalies need to focus on monitoring data, it is willing to be a sign that the terrorists carry out terrorist activities, which also provides technical support for our prediction and prevention of terrorist and criminal activities.

After analyzing all the data, we have come to the specific method how to evaluate human terror risk index and search suspects deliberate behavior. Finally, we wrote a letter to US President Barack Obama, the letter to the model we put forward on the basis of previous research, we recommend that the president should set the perfect terrorist risk index evaluation system, and to detect anomalies terrorist activity data, its strike, to prevent the tragedy from happening again.

2. General Assumptions

- We collected terrorist character of information and data related to terrorist attacks through public reports and databases.
- We are continuously monitoring the terrorists, the situation that information isn't collected does not exist.
- We don't consider the encryption information and don't take the special data into account.
- Some of the data is held in our hands can be clearly demarcate Category, the situation that the data is too ambiguous to classify isn't exist.
- The terrorist information is true, this model does not identify the authenticity of the data.

3. Symbol Description

u, v are any two nodes of the graph theory.

$r(u, v)$ is a binary variable that represents whether there is a link between the node u and the node v .

σ_{ij} is the number of shortest paths between i and j .

$\sigma_{ij}(K)$ is the shortest path between i and j through number of nodes K .

A is a $n \times n$'s adjacency matrix.

α is the proportional parameter.

β is the attenuation parameter.

N is the number of nodes in the network size and graph theory.

R is the relational matrix.

$\zeta(u, v)(W)$ is the dependent factor node u reach any other node by node v .

$occurrence(u, v)$ is the number of node by node u to communicate with other nodes v .

$path(u, w)$ is the number of the shortest distance between the node u and the node w .

$d(u, v)$ is the distance between node u and node v .

k_i is the number of nodes adjacent nodes.

e_{ij} is the nearest paths between node i and node j

$AT_{i,T}$ is the matrix values of the members i of the task T .

$|T|$ is the total number of tasks that the network members undertake.

W_r is the standard score of the monitored people get in religion

W_n is the standard score of the monitored people get in name

W_a is the standard score of the monitored people get in age

W_g is the standard score of the monitored people get in gender

W_m is the standard score of the monitored people get in marriage

W_{na} is the standard score of the monitored people get in nationality

W_e is the standard score of the monitored people get in duration

W_j is the standard score of the monitored people get in job

W_c is the standard score of the monitored people get in criminal history

W_s is the standard score of the monitored people get in social presence

W_{ab} is the standard score of the monitored people get in recruiting ability

4. Risk Index

4.1 Traditional Evaluation Model

4.1.1 The TREI Model

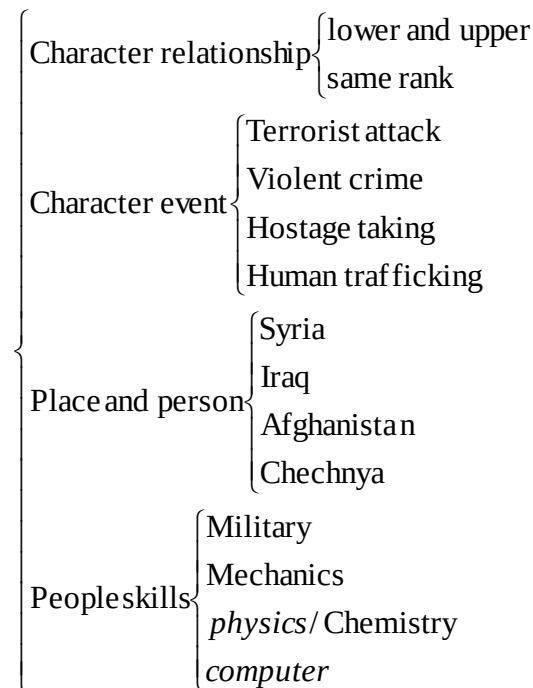
Network terrorist threat index of the monitored people can be solve by the following set of equations:

Relation Attribute	$\left\{ \begin{array}{l} \text{Degree} \quad C_d(u) = \sum r(u, v) \\ \text{Closeness} \quad C_c(u) = \frac{1}{\sum_{v \in V}(u, v)} \\ \text{Betweenness} \quad C_B(K) = \sum_{i \neq j \neq k \in V} \frac{\sigma_{ij}(K)}{\sigma_{ij}} \\ \text{Eigenvector} \quad x_i = \mu \sum_{j=1}^n \alpha_{ij} x_j \\ \text{Bonacich Power} \quad c(\alpha, \beta) = \alpha(I - \beta R)^{-1} R 1 \\ \text{Network Efficiency} \quad C_{eff} = E(G) = \frac{1}{N(N-1)} \sum_{i \neq j \in G} \frac{1}{d_{ij}} \quad (C_{eff} E \in [0,1]) \\ \text{Dependence Centrality} \quad \zeta(u, v)(W) = \frac{\text{occurrence}(u, v)}{d(u, v) \times \text{path}(u, w)} \\ C_{dep(u, v)} = 1 + \sum_{w=1}^n \zeta(u, v)(W), u \neq v \neq w \\ \text{Clustering Coefficient} \quad C_i = \frac{2 \{e_{ij}\} }{k_i(k_i - 1)} \\ \text{Information Centrality} \quad C_I(n_i) = \frac{1}{c_{ii} + (T - 2R)/g} \end{array} \right.$
Task Attribute	$\left\{ \begin{array}{l} \text{Cognitive Distinctiveness} \quad CD_{ij} = \frac{\sum_{T=1}^{ T } (AT_{i,T} * \sim AT_{j,T}) + (\sim AT_{i,T} * AT_{j,T})}{ T }, \\ CD_{i,i} = 0 \\ \text{Cognitive Expertise} \quad CE_{ij} = \frac{\sum_{T=1}^{ T } (\sim AT_{i,T} * AT_{j,T})}{(T - \sum_{T=1}^{ T } AT_{i,T})}, \quad CE_{i,i} = 0 \\ \text{Cognitive Resemblance} \quad CT_{i,j} = \frac{\sum_{T=1}^{ T } (AT_{i,T} * AT_{j,T}) + (\sim AT_{i,T} * \sim AT_{j,T})}{ T }, \\ CD_{i,i} = 0 \\ \text{Task Exclusivity} \quad TEI = \sum_{j=1}^{ T } AT(i, j) * \exp(1 - \text{sum}(AT(ij))) \\ \text{Socio Economic Power} \quad \text{matrix } N = [AK \ AR \ AT], SEP \text{ is equal to the Line} \\ \text{Degree of matrix } N \end{array} \right.$
Resource Attribute	$\left\{ \begin{array}{l} \text{Cognitive Distinctiveness} \quad CD_{ij} = \frac{\sum_{R=1}^{ R } (AR_{i,R} * \sim AR_{j,R}) + (\sim AR_{i,R} * AR_{j,R})}{ R }, \\ CD_{i,i} = 0 \\ \text{Cognitive Expertise} \quad CE_{ij} = \frac{\sum_{R=1}^{ R } (\sim AR_{i,R} * AR_{j,R})}{(R - \sum_{R=1}^{ R } AR_{i,R})}, \quad CE_{i,i} = 0 \\ \text{Cognitive Resemblance} \quad CR_{i,j} = \frac{\sum_{R=1}^{ R } (AR_{i,R} * AR_{j,R}) + (\sim AR_{i,R} * \sim AR_{j,R})}{ R }, \\ CD_{i,i} = 0 \\ \text{Task Exclusivity} \quad REI = \sum_{j=1}^{ R } AR(i, j) * \exp(1 - \text{sum}(AR(ij))) \\ \text{Socio Economic Power} \quad \text{matrix } N = [AK \ AR \ AT], SEP \text{ is equal to the Line} \\ \text{Degree of matrix } N \end{array} \right.$

$$\text{Skill Attribute} \left\{ \begin{array}{ll}
 \text{Cognitive Distinctiveness} & CD_{ij} = \frac{\sum_{K=1}^{|K|} (AK_{i,K} * \sim AK_{j,K}) + (\sim AK_{i,K} * AK_{j,K})}{|K|}, \\
 & CD_{i,i} = 0 \\
 \text{Cognitive Expertise} & CE_{ij} = \frac{\sum_{K=1}^{|K|} (\sim AK_{i,K} * AK_{j,K})}{(|K| - \sum_{K=1}^{|K|} AK_{i,K})}, \quad CE_{i,i} = 0 \\
 \text{Cognitive Resemblance} & CK_{i,j} = \frac{\sum_{K=1}^{|K|} (AK_{i,K} * AK_{j,K}) + (\sim AK_{i,K} * \sim AK_{j,K})}{|K|} \\
 & CK_{i,i} = 0 \\
 \text{Task Exclusivity} & KEI = \sum_{j=1}^{|K|} AK(i,j) * \exp(1 - \text{sum}(AK(ij))) \\
 \text{Socio Economic Power} & \text{matrix } N = [AK \ AR \ AT], \text{ SEP is equal to the Line Degree of matrix } N
 \end{array} \right.$$

4.1.2 Model Illustration

Network behavior can be divided into general situation, access to the site, send and receive e-mail, upload and download, instant messaging, chat, forums, online games, streaming video and Telnet. We can't get to know the communication and email content of the monitored people, but we can get the relationship between them. Each person can be regarded as a node, if there is a link between any two people, then these two nodes will have a way, the final link between the people and the formation of A. We can think of this person and the terrorist event if the person is to have participated in some of the terrorist activities. Similarly, if a person is under the control of a terrorist attack skill and a certain kind of weapon, we also have reason to think that they have a connection with this information



Network properties can be divided into four categories, by the above equations

were calculated its index score, we can evaluate its degree of risk. Different network properties can be divided into four categories altogether, relationship property, task attributes, skills, attributes and resource properties. For different properties, we need to build a different relationship matrix.

i) relationship property, build $n \times n$ relation matrix $G(A)$, where $a_{ii} = 0$, n represents the number of terrorists.

$$\begin{cases} A_{ij} = 1, & \text{if terrorist } i \text{ and } j \text{ have relation} \\ A_{ij} = 0, & \text{if terrorist } i \text{ and } j \text{ have no relation} \end{cases}$$

ii) People – task, build $n \times m$ relation matrix $G(M)$, where, m represents the number of task.

$$\begin{cases} E_{ij} = 1, & \text{if a terrorist undertaken a task} \\ E_{ij} = 0, & \text{if a terrorist not undertaken a task} \end{cases}$$

iii) People-resource, build $n \times r$ relation matrix $G(R)$, r represents a kind of resource

$$\begin{cases} R_{ij} = 1, & \text{if a terrorist has a resource} \\ R_{ij} = 0, & \text{if a terrorist not has a resource} \end{cases}$$

iiii) People-skill, build $n \times k$ relation matrix $G(K)$, where, k represents total skill in network.

$$\begin{cases} R_{ij} = 1, & \text{if a terrorist has a skill} \\ R_{ij} = 0, & \text{if a terrorist has no skill} \end{cases}$$

After the matrix is built, applied social network analysis software Ucient, you can get terrorist network model.

4.1.3 Model Application

The evening of November 2015 13, terrorists attacked Paris, then Paris police arrested five suspects, they are Bilal Hadfi, Samy Amimour, Ahmed Almuhammed, Abdel-Hamid Abu Oud, Abdeslam Salah.

In order to verify the feasibility of TREI model, we can calculate the terror risk index of five suspects who have already arrested. Because we cannot get the real communication, relationships between them, so when calculating their terror risk index for their virtual network attributes. According to the model, establish the following four relationship matrix, for convenience, using the short form BH, SA, AA, AA and AS on behalf of the above-mentioned five suspects.

● People Relationship Matrix

Table 1

	BH	SA	AA	AAO	AS
BH	0	0	1	1	0
SA	0	0	0	1	1
AA	1	0	0	0	1

AAO	1	1	0	0	1
AS	0	1	1	1	0

- People - time matrix Table 2

	I_1	I_2	I_3	I_4	I_5	I_6
BH	1	1	0	1	1	0
SA	0	1	0	1	0	1
AA	1	1	1	0	0	1
AAO	0	0	0	1	1	0
AS	1	0	0	0	0	1

where, I_i represents the i -th event, $i = 1, 2, 3, 4, 5, 6$.

- People - place matrix Table 3

	P_1	P_2	P_3	P_4	P_5
BH	0	0	0	1	1
SA	1	1	0	1	0
AA	0	0	1	1	1
AAO	1	0	1	0	0
AS	1	0	1	1	1

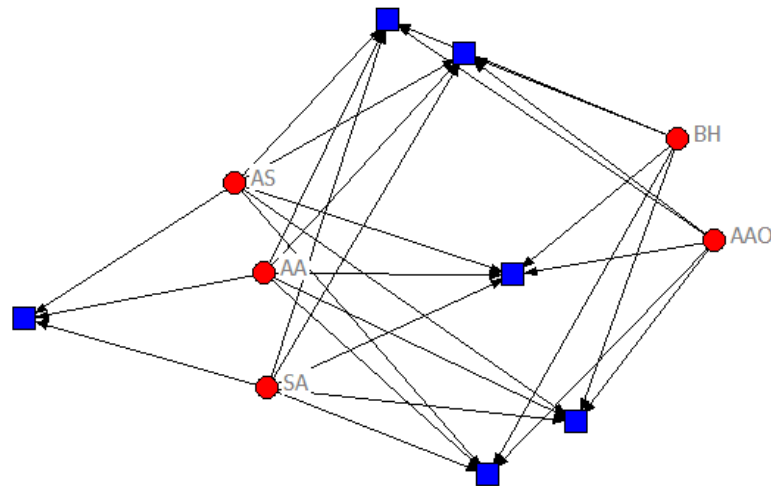
where, P_i represents the i -th place, $i = 1, 2, 3, 4, 5$.

- People - skill matrix Table 4

	I_1	I_2	I_3	I_4	I_5	I_6	I_7
BH	1	1	1	1	0	0	1
SA	1	0	1	1	0	1	1
AA	1	0	0	0	1	0	0
AAO	0	1	0	1	1	1	0
AS	1	0	1	1	1	0	0

Where, S_i represents the i -th skill, $i = 1, 2, 3, 4, 5, 6, 7$.

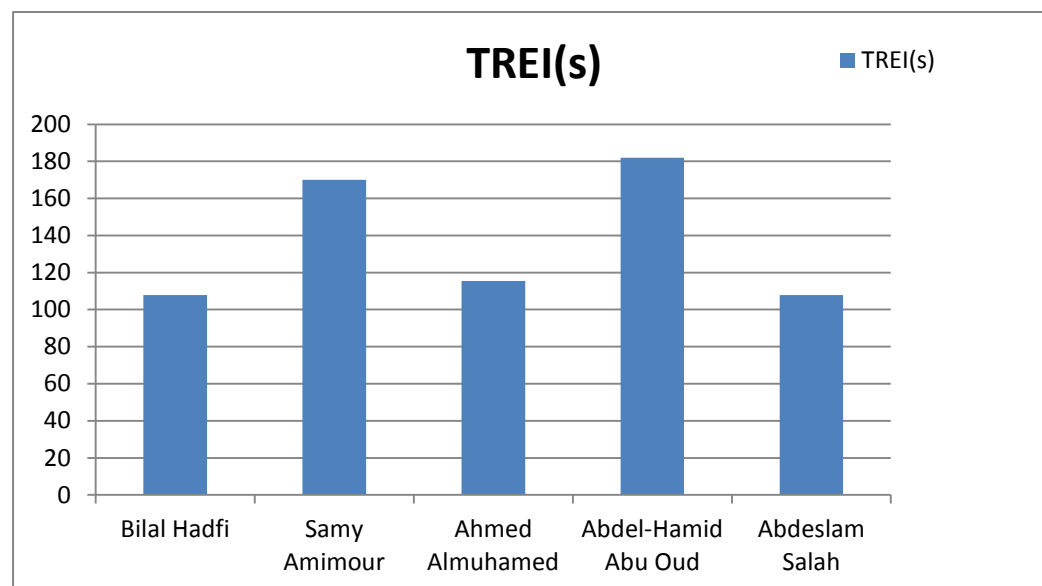
The relationship matrix into the social network analysis software Unicet6.0 for analysis, the following results can be obtained.



Five terrorists on four categories of network attributes are calculated, which is finishing up convergence, you can get five criminals Network Properties score.

Table 5

	Relation Attribute	Task Attribute	Skill Attribute	Resource Attribute	Total	Ranking
BH	39.48	35.90	14.57	7.61	97.56	5
SA	39.23	93.56	12.98	12.63	158.40	2
AA	50.65	32.48	13.11	7.61	103.85	3
AAO	64.13	83.82	6.78	15.32	170.05	1
AS	43.61	30.13	17.23	7.59	98.50	4



From the calculation results, Abdel-Hamid Abu Oud has the highest network properties score, in the first place, and he is The Paris bombing mastermind terrorist attacks. Ranked second was Samy Amimour, who is the prime suspect, and his name has also been included in the list of internationally wanted persons. The three others left, Bilal Hadfi, Ahmed Almuhammed and Abdeslam Salah have same score, around 100, and they all participated in the Paris bombings.

The above examples illustrate that the model we established to predict individual

terror risk index can predict the people we monitored terror risk index accurately, and help detect those who make timely judgments and plans, in order to prevent large-scale the occurrence of violent and terrorist incidents.

4.2 Expanded Model

4.2 .1 The TREI(S) Model

TREI is a basic model for predicting the terrorism risk index. However , this basic model cannot evaluate the risk index completely. In order to make the model more perfect, we have to take other important factors into consideration. We mainly consider,

- 1) The social attribute of the monitored people, such as name (whether or not wanted), age, gender, marriage situation, nationality, education, religion ,history of crime ,recruiting ability, etc.
- 2) According to the references we have found, we classify the projects, and marked them.

The tables below show classifications and its weights.

Table 6

Attribute name	Possibility	Weight
religion	A	9
	B	5
	C	2
	No religion	1

Based on past experience of terrorist attacks, can be found, the terrorists often carry out terrorist activities in the name of religion. Compared to people without faith, religious fanatics are often more threatening. Meanwhile, in the specific terrorist incidents, such as terrorist attacks caused by religious conflict, terrorists tend to have a common religion, in this case, people who believe in a particular religion, compared to people of other religions, tend to be more It may become a terrorist. Therefore, for a terrorist event, we can say that a religious belief (for example A) people more threatening, more likely to become a key figure in the network, therefore, religion A, we give it the highest score here 4 points. For other than A, B or C religion, we give it low scores were 3 or 2 points. Without faith that is considered the least threat, therefore giving it the lowest score 1 point.

Table 7

Attribute name	Possibility	Weight
	In international data base	9
name	Terrorists are known to exist in the Region A	8

B	7
C	6
Other countries	2

If a terrorist's name appears in the archives of international terrorism, then he can be considered threatening higher compared to other terrorists, therefore their assigned 9; secondly to appear on the terrorism file a country, select the country standard occurrence and degree of harm based on the country's terrorist activities. According to a similar theory, assign to each of the following features. You will see other assignment criteria in appendix.(Tab. 8 to Tab. 16)

The traditional model TREI should be modified , and the improved TREI(s) model can be written as

$$SW = \frac{W_r}{4} + \frac{W_n}{5} + \frac{W_a}{3} + \frac{W_g}{2} + \frac{W_m}{2} + \frac{W_{na}}{4} + \frac{W_e}{6} + \frac{W_j}{4} + \frac{W_c}{4} + \frac{W_s}{4} + \frac{W_{ab}}{4}$$

4.2.2 Model Application

Five terrorists' information who had taken part in this terrorism attacks can be collected online. Assigned the value of the characteristics of them and calculates the SW(Tab. 17)

Table 17

	name	gender	age	marriage	education	nationality	job	Criminal history	Social outlook	recruiting ability	SW
BH	8	2	1	2	5	1	4	8	7	2	10.26667
SA	9	2	3	2	2	1	4	9	8	4	11.63333
AA	8	2	3	2	2	4	4	9	7	3	11.68333
AAO	8	2	3	2	5	1	4	9	8	4	11.93333
AS	8	2	3	1	2	1	4	8	4	3	9.43333

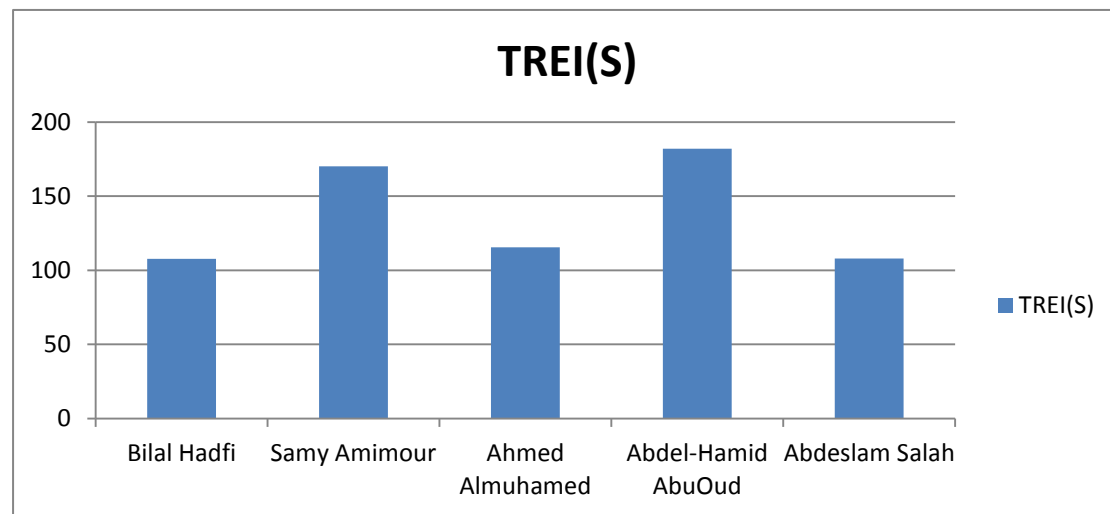
The social attribute scores and network attributes combination score can be obtained, and we will get five terrorists' terrorism index (Tab. 18)

Table 18

	SW	network properties score	Total	Ranking
Bilal Hadfi	10.26667	97.56	107.82667	4
Samy Amimour	11.63333	158.40	170.03333	2
Ahmed Almuhammed	11.68333	103.85	115.53333	3

Abdel-Hamid AbuOud	11.93333	170.05	181.98333	1
Abdeslam Salah	9.433333	98.50	107.93333	5

Table 19



From the result above, Abdel-Hamid Abu Oud has the highest TREI(s) score, second to him is Samy Amimour, and Abdeslam Salah has the lowest score. Otherwise, Bilal Hadfi gets the lowest score. Apparently, the TREI(S) model can be more accurately and comprehensively reflects the individual terrorist threat index. When we add the community property factors, this person's upbringing, personal experiences and his commitment to social responsibility are taken into account, it is possible to assess the full range of multi-angle individual terror threat index, and a more accurate assessment be made.

5. Big data classification and outlier handling

5.1 Traditonal model

5.1.1K- neighborhood outlier detection algorithm

1) Simulation algorithm1.0

When we get a lot of being monitored by the data, these data need to be fast and efficient approach to classification. In order to dig out the terrorists' information in large amounts of data.^[7] We need to separate out the abnormal data and process these data. And the parted data may hide terrorist plans to walk the next. Here, we applied K- neighborhood outlier detection algorithm.

Based on the analysis above, we design our simulation arithmetic as follows.

The calculation processes are as follows:^[8]

- **Step1:** Calculation dataset all objects except the distance to the object P to P.

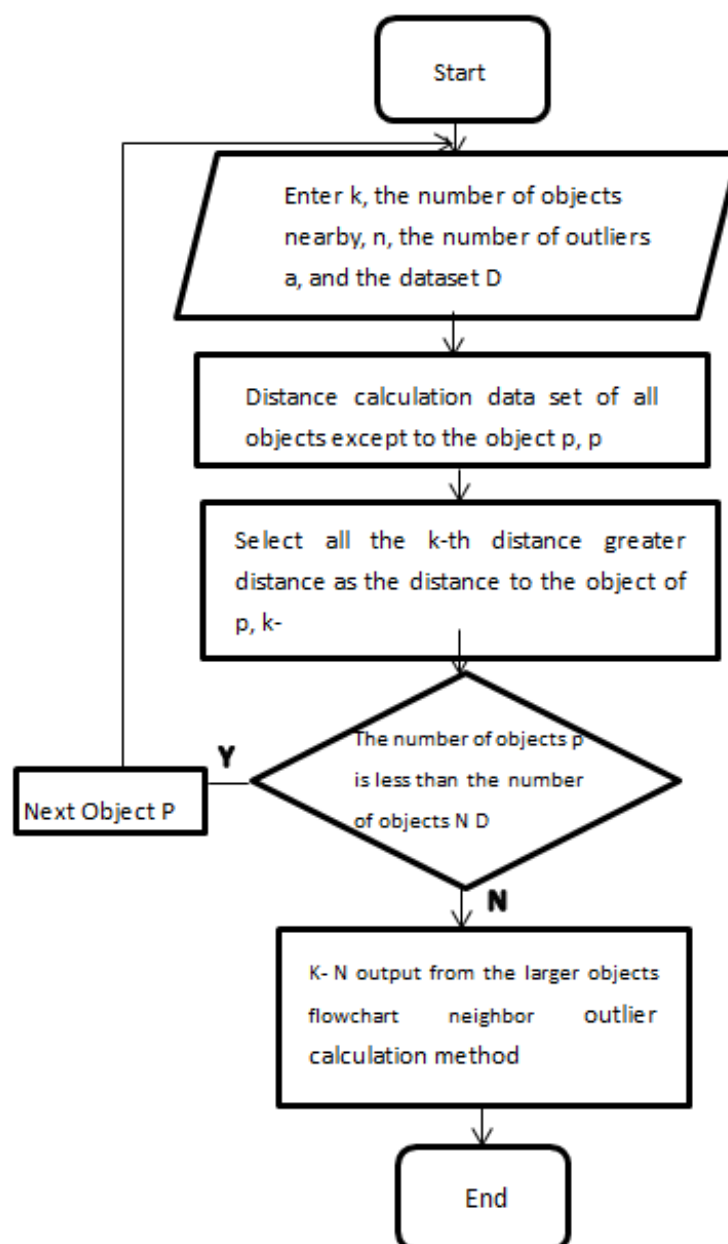
- **Step2:** Select k smallest distance.
- **Step3:** Select the k largest distance as the distance from the target of P k-.

The main steps of the algorithm are as follows:

- **Step1:** Enter: (1) d-dimensional data set D, there are data of N(2) the number of data objects neighbor
- **Step2:** Outliers number of points n Output: true outlier status
- **Step3:** According to object properties and spatial relationships, determine each object k. distance
- **Step4:** The k- distance sort descending
- **Step5:** k- larger distance between the front of an object is considered to be outliers^{[7][9]}

2) Flow chart

The figure below is the flow chart of the^[9]



3) Solution

In order to verify the feasibility of our model, we run the program and run the results. Because the real information of the monitored person cannot be obtained, we make some virtual data.(Tab.21) The data in Table indicates that the classification of the database has been classified, and the following classification process is based on this. Table is the expression big data that we simulate.

Table 20 Data in database

a	1	2	3	4	5	6	7	8	9	10	11
b	10	9	8	7	6	5	4	3	2	1	5
c	3	3	3	3	3	3	3	3	3	3	8
d	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3	-3
e	1	2	3	4	4	5	5	6	6	7	4
f	7	7	6	6	5	4	4	3	2	1	2
g	8	8	7	7	6	3	3	3	2	4	3
h	0	0	1	1	-2	5	7	8	9	15	-5

Table 21 Virtual data

1	2	3	2	4	2	3	5	6	7	5
2	3	4	2	1	-5	-8	15	12	3	4
8	7	2	3	5	2	3	5	0	-8	5
-3	-2	2	4	0	2	-5	5	0	2	8
-4	-4	-4	-4	-4	7	5	0	-2	1	2
1	2	3	4	4	4	-8	1	2	3	5
4	4	3	2	1	-7	-5	-2	-3	0	2
0	0	1	1	-2	-5	4	5	0	5	-1

Table 22 Result

a	1	2	3	2	4	2	3	5	6	7	5
a	2	3	4	2	1	-5	-8	15	12	3	4
b	8	7	2	3	5	2	3	5	0	-8	5
c	-3	-2	2	4	0	2	-5	5	0	2	8
d	-4	-4	-4	-4	-4	7	5	0	-2	1	2
c	1	2	3	4	4	4	-8	1	2	3	5
c	4	4	3	2	1	-7	-5	-2	-1	0	2
c	3	3	3	2	4	-7	-5	-2	-3	0	2
c	0	0	1	1	-2	-5	4	5	0	5	-1

5.1.2 Analysis of the Result

Because of the variety of network information, the data is huge, and the resources are limited. we simulate the network data, and then describe the information as a feature vector. For example, we use the letter 'a' to represent the "mail class" information, and the "time" (or "IP address", etc.) to send the message to the feature vector.

As shown in the appendix 2, this is the simulation implementation of our

classification algorithm1.0.

In this place, we have carried out the appropriate abstraction of the terrorists' information, and describe the information characteristics of each kind of information with its corresponding feature vector, and the correlation between them.

As to Test data in Table 21, each row header letter said a category of information, a feature vector composed of all subsequent figures, the vector describes a belongs to the category of information characteristics. That is, a message corresponds to a category.

In Table, each line represents a message, and the classification results are saved in the Result (Table 22) by using the algorithm of the model.

5.1 Improved model

5.1.1 SKNN Algorithm

1) Improved basis

The traditional KNN method has two obvious disadvantages: the sample size calculation is large and the sample library is strong. The research of this model is to deal with the big data of the Internet, so it is necessary to reduce the amount of sample similarity, improve the classification speed of KNN, so as to improve the practical application of KNN in information classification. The improved method of reducing sample similarity can be divided into 3 categories:^[11]

- Reduce the computational quantity of sample similarity by reducing the dimension of the feature space of the sample. If the number of feature words is too big, it will lead to the high dimension of information vector, increasing the computational cost, and too many dimensions to distinguish ability.
- Use a small sample to reduce the amount of sample similarity. This method is used in the original training sample to select some representative samples as the new training sample, or delete some samples of the original training sample set, the remaining sample as the new training sample, so as to reduce the training sample set. This method mainly has the condensing Hart algorithm, editing Devijver algorithm and MultiEdit Wilso algorithm. Kuncheva using genetic algorithm in this area also carried out some research.
- Improve the speed of the KNN algorithm to find the nearest neighbor of K. This method is to find the nearest neighbor of the sample to be classified in the shortest possible time by the fast search algorithm.

The main idea of this paper is to sort the sample in order to search the nearest neighbor of K in order to reduce the number of K SKNN. Therefore, the SKNN algorithm is first to determine a reference point R, according to the distance of the sample to R, and establish an index table, and then to the X R, then find the range of X to Q, and then find the R, K, and then find Q, the Q is the center of the sample, and then the K sample is used to search for the nearest neighbor Q. At this point to find the exact x k nearest neighbor.

2) Establish the ordered linear space of the training sample library:

- **Step1:** randomly selects a sample as a reference point R ($R_1, R_2, R_3, \dots, R_n$),

n is the dimension of the sample feature vector.

- **Step2:** according to the formula (the distance between the sample I and sample n ; J is the total number of attributes, is the I of the sample h attribute; is the weight of the H attribute. Each sample to the R of the distance D , using the insertion sort method into an orderly queue queue, each node including the corresponding sample to the distance $R D$, the category and feature vector. In order to consider the time cost of reading the disk, the index table of the training sample is set up in order to consider the time cost of the disk
- **Step3:** Only the $1, 1 + L, 1 + 2L$, in the ordered sample queue in the index table, $\dots, 1 + iL \dots (1 \leq i \leq [m/L])$ the location of the sample in the ordered queue and the distance to the R .

5.1.2 Results and Analysis

By improved KNN algorithm-SKNN to classified eight hundred sample data, and we get the results below,

Table 23

	SKNN Classification time	KNN Classification time	Whether the k nearest neighbor is the same that SKNN and KNN find
$k=6$	16s	6min25s	The same
$k=10$	19s	6min28s	The same
$k=20$	27s	6min39s	The same
$k=40$	47s	7min12s	The same

From the table, we can see that with the same number of samples, SKNN can categorize these number in shorter time, which is 24 times faster than the traditional model SKNN.

6. Strength and Weakness

Strength:

- We improved the traditional risk index model(TREI) and transformed it into the TREI(S) model, which consider more factors and is thus better suited to the evaluation system.
- We divided the factors into two parts, social properties and network one. Social properties include 11 elements, and the network one has 4 attributes.
- We applied KNN algorithm into the big data categorization, which is a effective, fast and automatic way to categorize the monitoring data.

Weakness

- Due to difficulties in gathering data, the data analysis may result in a margin of error value and the actual value .

7. Letter to President Obama

Dear President Obama,

The occurrence of '9-11' incident, profoundly changed the international relations, changed the country's foreign strategy, also changed people's social life. This incident shows that terrorism has constituted a global threat to peace and stability. In the face of such a situation, each country is duty bound, can only work together, gather everything in its power to combat and eradicate terrorism. There is no doubt that, in this struggle, the scientific and technological power is very important. Especially in network security, it is urgent to strengthen the management, closely monitor the propaganda of terrorism, the possible existence of terrorists and other acts involving terrorism.

However, the network data is various, and the number is huge. And different from the traditional network, the terrorist network does not have a strict hierarchical structure, the members of the contact between the random and covert, making the network relationship of terrorist networks is complex and diverse, how to measure the possibility of network users to conduct terrorist acts (that is, the body of the threat index), how to effectively classify big data into the current difficulties. For this reason, my partner and me did a series of research and developed a set of strict, regular and efficient big data processing method on the individual threat index, the accurate classification of network data, effective treatment and other technologies. And according to the economic, political, cultural and other factors of the terrorist may appear, more intensive areas were predicted. I hope our research can provide some basis for the decision of your policy.

In view of the above problems, we use the method of qualitative analysis and quantitative analysis to analyze the combination of qualitative analysis and quantitative analysis,

area	Middle East	North Africa	South East Asia	South Asia	Eastern Europe
Individual threat index	181.98	122.6	95.4	80.5	54.2

Obviously, the individual threat index in the top two in the Middle East and North Africa region is fully in line with the reality, so the accuracy of this result can be achieved in the actual requirements, and the subsequent South East Asia, South Asia is the possibility of our personal growth of the breeding ground for terrorism. To this end, we suggest that these areas to strengthen the safety of prevention and control, to prevent the spread of terrorism.

In counter-terrorism operations to eliminate the root of terrorism survival is anti-terrorism stopgap fundamental decision, but from the point of view of the current situation, the survival and development of terrorism by the influence of many factors, such as political equality and economic backwardness leads to poverty, culture,

religion, revenge psychological. These problems cannot be eliminated in the short term. In this case, it is more important to prevent the attack when it is unable to eliminate the root causes. To prevent and combat the combined strategy is good way to control the crime of terrorism.

First, the establishment of a sound anti-terrorism security law, to protect the United States and the United States to develop a comprehensive law, for the government and the whole society to counter terrorism, safeguard national security measures to provide legal basis. Bill should be given the relevant government departments in domestic terrorism and safeguard national security in a wide range of power, provisions of the responsibilities and obligations of the U.S. government, private organizations and people in the fight against terrorism and the maintenance of security, authorized by the United States to carry out large-scale understanding, tracking and interception in the communication activities related to country of foreigners and citizens to view public telephone and Internet records, personal letters and e-mail, view the domestic library readers borrow books and bookstores, the individual shop book records, secret search suspect residence and so on.

As the research goes on, the complex network theory should make a difference in predicting the network of terrorist organizations, especially in the prediction of the spatial and temporal evolution of the network of the major terrorist organization.

The application of network science in the war against terrorism has broad prospects. At present, the research is facing the new challenges in the era of big data, including data collection, processing, the time and space evolution of terrorist organization, network dynamics, and the control of its network. However, with the development of big data and information technology, big data science and network science need to be closely integrated; the urgent need to drive innovation method. At present, the method and means of monitoring terrorist organization is getting better and better, and the open source intelligence is growing. The United States should establish the relevant anti-terrorism database actively, make big data source more timely, reliable and accurate.

At the same time, it should be recognized that, in addition to exploring the network theory of terrorism, the United States is not a simple way to solve the problem of war, which is a comprehensive security issues involving the network space, it is involved in the network space, should pay attention to both hands, the implementation of comprehensive management, specimen and governance, terrorism, the United States should oppose double standards, promote international cooperation and play a role in the United nations.

In addition, should strive to counter-terrorism operations more transparent, efforts to form a 'people's war' in the ocean and the masses of the ideological and political situation, to promote the construction of a safe and harmonious world.

**Best regards,
Team #2125.**

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9. Appendix

Table 8

Attribute name	Possibility	Weight
age	15-25	1
	25-40	3
	40-	2

Table 9

Attribute name	Possibility	Weight
gender	Male	2

female	1
--------	---

Table 10

Attribute name	Possibility	Weight
marriage	married	2
	unmarried	1

Table 11

Attribute name	Possibility	Weight
nationality	Syria	4
	Iraq	3
	Afghanistan	2
	Other countries	1

Table 12

Attribute name	Possibility	Weight
education	Bachelor degree and above /Reading	9
	Technical secondary school / junior college	7
	Senior high school	5
	Junior high school	5
	Primary school	2
	No admission	2

T Table 13

Attribute name	Possibility	Weight
job	Military	9
	Chemical/ Physics /Machinery	8
	Computer	7
	Else	4

Table 14

Attribute name	Possibility	Weight
Criminal history	Severity	9
	Moderate	8
	Mild	5
	None	3

Table 15

Attribute name	Possibility	Weight
Social outlook	Very poor	8
	Poor	7
	General	4
	Good	1

Table 16

Attribute name	Possibility	Weight
	Strong	4
Recruiting	General	3
ability	Weak	2
	Little	1

2)

```
#include <iostream>
#include <string>
#include <vector>
#include <set>
#include <map>
#include <fstream>
#include <sstream>
#include <cassert>
#include <cmath>
#include <windows.h>
#include <cstdio>
using namespace std;
struct sample
{
    string type;
    vector<double> features;
};
struct typeDistance
{
    string type;
    double distance;
};

bool operator < (const typeDistance& lhs, const typeDistance& rhs)
{
    return lhs.distance < rhs.distance;
}

void readTrain(vector<sample>& train, const string& file)
{
    ifstream fin(file.c_str());
    if (!fin)
    {
        cerr << "File error!" << endl;
        exit(1);
    }
    string line;
    double d = 0.0;
```

```
while (getline(fin, line))
{
    istringstream sin(line);
    sample ts;
    sin >> ts.type;
    while (sin >> d)
    {
        ts.features.push_back(d);
    }
    train.push_back(ts);
}
fin.close();
}

void readTest(vector<sample>& test, const string& file)
{
    ifstream fin(file.c_str());
    if (!fin)
    {
        cerr << "File error!" << endl;
        exit(1);
    }
    double d = 0.0;
    string line;
    while (getline(fin, line))
    {
        istringstream sin(line);
        sample ts;
        while (sin >> d)
        {
            ts.features.push_back(d);
        }
        test.push_back(ts);
    }
    fin.close();
}

double euclideanDistance(const vector<double>& v1, const vector<double>& v2)
{
    assert(v1.size() == v2.size());
    double ret = 0.0;

    for (vector<double>::size_type i = 0; i != v1.size(); ++i)
    {
        ret += (v1[i] - v2[i]) * (v1[i] - v2[i]);
    }
}
```

```
    }
    return sqrt(ret);
}

void initDistanceMatrix(vector<vector<double> >& dm, const vector<sample>&
train, const vector<sample>& test)
{
    for (vector<sample>::size_type i = 0; i != test.size(); ++i)
    {
        vector<double> vd;
        for (vector<sample>::size_type j = 0; j != train.size(); ++j)
        {
            vd.push_back(euclideanDistance(test[i].features, train[j].features));
        }
        dm.push_back(vd);
    }
}

void knnProcess(vector<sample>& test, const vector<sample>& train, const
vector<vector<double> >& dm, unsigned int k)
{
    for (vector<sample>::size_type i = 0; i != test.size(); ++i)
    {
        multimap<double, string> dts;
        for (vector<double>::size_type j = 0; j != dm[i].size(); ++j)
        {
            if (dts.size() < k)
            {
                dts.insert(make_pair(dm[i][j], train[j].type));
            }
            else
            {
                multimap<double, string>::iterator it = dts.end();
                --it;
                if (dm[i][j] < it->first)
                {
                    dts.erase(it);
                    dts.insert(make_pair(dm[i][j], train[j].type));
                }
            }
        }
        map<string, double> tds;
        string type = "";
        double weight = 0.0;
```

```
        for (multimap<double, string>::const_iterator cit = dts.begin(); cit !=
dts.end(); ++cit)
        {
            tds[cit->second] += 1.0 / cit->first;
            if (tds[cit->second] > weight)
            {
                weight = tds[cit->second];
                type = cit->second;
            }
        }
        test[i].type = type;
    }
}

void writeTest(const vector<sample>& test, const string& file)
{
    ofstream fout(file.c_str());
    if (!fout)
    {
        cerr << "File error!" << endl;
        exit(1);
    }
    for (vector<sample>::size_type i = 0; i != test.size(); ++i)
    {
        fout << test[i].type << '\t';
        for (vector<double>::size_type j = 0; j != test[i].features.size(); ++j)
        {
            fout << test[i].features[j]<<" ";
        }
        fout << endl;
    }
}

void knn(const string& file1, const string& file2, const string& file3, int k)
{
    vector<sample> train, test;
    readTrain(train, file1.c_str());
    readTest(test, file2.c_str());
    vector<vector<double>> dm;
    initDistanceMatrix(dm, train, test);
    knnProcess(test, train, dm, k);
    writeTest(test, file3.c_str());
}
```

```
int main()
{
    knn("train.txt", "test.txt", "result.txt", 11);
    return 0;
}
```